OPTICAL COHERENCE TOMOGRAPHY ANGIOGRAPHY OF THE MACULA AFTER PLAQUE RADIOTHERAPY OF CHOROIDAL MELANOMA

Comparison of Irradiated Versus Nonirradiated Eyes in 65 Patients

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Purpose: To study radiation retinopathy after plaque radiotherapy of choroidal melanoma using optical coherence tomography angiography.

Methods: Retrospective comparative analysis of 65 consecutive patients with choroidal melanoma, treated with standard dose I-125 plaque radiotherapy and imaged with optical coherence tomography angiography. A comparison of irradiated versus contralateral, nonirradiated (control) eyes was performed.

Results: The mean patient age was 55 years. Underlying medical diseases included diabetes mellitus (4/65, 4%) or hypertension (25/65, 38%), but no patient demonstrated disease-related retinopathy. The mean pretreatment melanoma diameter was 11 mm and mean thickness was 5 mm. The mean radiation dose to the foveola was 5663 centiGray. At mean follow-up of 46 months after plaque radiotherapy, the most frequent qualitative finding on optical coherence tomography angiography (irradiated eye) was nonperfusion in the superficial capillary plexus (19/65, 29%) and deep capillary plexus (20/65, 31%), followed by loss of choriocapillaris within tumor margins (11/65, 17%). The quantitative findings revealed foveal avascular zone with significantly larger mean area (irradiated vs. nonirradiated eye) in the superficial plexus (0.961 vs. 0.280 mm^2, $P < 0.0001$) and deep plexus (1.396 vs. 0.458 mm^2, $P < 0.0001$), even in eyes without clinical evidence of radiation maculopathy (superficial 0.278 mm^2, $P = 0.03$; deep 0.454 mm^2, $P = 0.02$). Parafoveal capillary density (superficial and deep) was decreased in all irradiated eyes ($P < 0.001$). This difference was maintained after subgroup analysis of eyes with ($P < 0.001$) or without ($P < 0.001$) clinical evidence of radiation maculopathy. Mean logMAR visual acuity was significantly reduced in irradiated eyes (0.7 vs. 0.1 [Snellen equivalent 20/100 vs. 20/25], $P < 0.001$) and the reduced vision was significant even in eyes without clinical evidence of radiation maculopathy (0.4 vs. 0.1 [Snellen equivalent 20/50 vs. 20/25], $P < 0.001$).

Conclusion: Optical coherence tomography angiography demonstrated significant enlargement of the foveal avascular zone and decreased parafoveal capillary density of both superficial and deep capillary plexuses in eyes after plaque radiotherapy of choroidal melanoma, even in eyes with no clinical evidence of radiation maculopathy.

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Choroidal melanoma is a life-threatening malignancy that is generally managed with enucleation of the globe or focal radiotherapy directed at the intraocular mass to achieve tumor control.[1–3] Focal radiotherapy is delivered using plaque brachytherapy or charged particle teletherapy. Both can result in short-term and long-term damage to the neurosensory retina and choroid, leading to vascular compromise with leakage, edema, hemorrhage, nonperfusion, and neovascularization.[4–8] This ultimately can result in permanent vision loss.[9]

In 1999, Gunduz et al[4] studied 1,300 consecutive patients with plaque-irradiated posterior uveal melanoma and reported 5-year Kaplan–Meier estimates of nonproliferative retinopathy in 42% and proliferative retinopathy in 8%. The most important factor specifically predictive of radiation maculopathy was tumor base within 4 mm of the foveola. The judgment regarding radiation retinopathy, in that series, was based on ophthalmoscopic evidence alone because optical coherence tomography (OCT) was not commercially available at that time. Later, in 2008, Horgan et al[6] used time domain OCT to study the onset of radiation macular edema in 135 patients after plaque radiotherapy for choroidal melanoma. The median radiation dose to the foveola was 3,406 centiGray (cGy) and they documented OCT-evidence of macular edema in 17% by 6 months, 40% by 1 year, and 61% by 2 years. Despite these early OCT findings, ophthalmoscopic evidence of maculopathy was detected much later, on average at 17 months and only in 38% of cases. This underscored the limitation of ophthalmoscopy for the detection of radiation-related maculopathy. Most recently, in 2015, Mashayekhi et al[10] found very early subclinical, trace macular edema, in eyes with melanoma, even before plaque radiotherapy, suggesting that tumor-related factors could also contribute to later macular edema after radiotherapy.

Optical coherence tomography angiography (OCTA) is a new method for imaging the vascular tissue in the macular region.[11–14] This novel technology provides high-resolution detail of the two major capillary levels in the parafoveal region including the superficial and deep capillary plexuses. This noninvasive vascular imaging modality is achieved without dye injection, using volumetric angiographic information and the principles of split-spectrum amplitude-decorrelation angiography (SSADA). Herein, we employ OCTA to study the macular region in eyes previously treated with plaque radiotherapy for choroidal melanoma.

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Materials and Methods

A retrospective chart review of all patients diagnosed with choroidal melanoma treated with plaque radiotherapy who had OCTA between February, 2015 and April, 2015 were included in this study. This study adhered to the tenets of the Declaration of Helsinki, and approval was obtained from the Institutional Review Board of Wills Eye Hospital. Informed consent was obtained from all subjects after explanation of the nature of the study. Data gathered included patient demographics, pretreatment tumor features, radiation parameters during plaque radiotherapy, postoperative clinical features at the time of OCTA, and specific OCTA features.

Patient demographics recorded included age, race (white, Hispanic, African-American), gender, medical history (diabetes mellitus, hypertension), and previous history of retinal disorders (diabetic retinopathy, hypertensive retinopathy). Baseline tumor features include largest basal diameter, tumor thickness, tumor location, distance to the optic nerve, distance to the foveola, and presence of subretinal fluid and cystoid macular edema. The foveola was defined as the center of the fovea based on fundus photography and OCT. The macular region was defined as the area within 3 mm from the foveola. Radiation parameters included radiation dose (cGy) to the tumor apex, tumor base, foveola, and optic disc. Subsequent treatments for tumor consolidation (transpupillary thermotherapy) and prevention of radiation side effects (intravitreal bevacizumab injection[15] and sector panretinal laser photocoagulation[16,17]) were documented. In addition, history or presence of radiation complications such as radiation maculopathy without macular edema, radiation maculopathy with macular edema (both cystoid and noncystoid), and radiation papillopathy were recorded. Clinically evident radiation maculopathy was documented by ophthalmoscopic examination and defined by the presence of retinal telangiectasia, microaneurysm, exudation, hemorrhage, edema, nerve fiber layer infarction, and neovascularization. Visual acuity at the study visit after plaque radiotherapy was also recorded.

Image Acquisition Protocol

The Optovue RTVue XR AVANTI, version 2014.2.0.13 (Optovue Inc, Fremont, CA) (awaiting Food and Drug administration approval, used under Institutional Review Board approval) was used to obtain all images in this study, with OCT scans obtained using a 840-nm wavelength laser with a bandwidth of 45 nm, 3-mm scan depth, and tissue

Table 1. Optical Coherence Tomography Angiography After Plaque Radiotherapy for Choroidal Melanoma (Patient Demographics)

Feature	Number (%), n = 65
Age, mean (median, range), years	55 (56, 12–81)
Race	
White	63 (98)
Hispanic	2 (2)
Sex	
Male	26 (40)
Female	39 (60)
Medical history	
Diabetes	4 (4)
Hypertension	25 (38)
Involved eye	
Right	33 (51)
Left	32 (49)

resolution of 5 μm. This technology can acquire 70,000 A-scans per second and 316 A-scans per B-scan. All patients required a standard 6-mm retinal map analysis protocol for volume scans. Central macular thickness (CMT) was recorded for analysis and senior investigators (C.L.S. and E.A.T.S.) judged for the presence of active cystoid macular edema. The same machine was also used for OCTA image acquisition. The scanning algorithm began with 2 B-scans captured at each fixed position before the next sampling location and there were two orthogonal OCTA volume scans (one horizontal and one vertical) used to minimize motion artifacts and fixation changes. Split-spectrum amplitude-decorrelation angiography algorithm was used to create OCTA en-face images. For maximum image detail, only 3-mm scans were performed with a scan area of 9 mm^2 centered on the foveola (parafoveal region) containing 304 × 304 pixels per en-face image. Image segmentation was performed using the built-in software (ReVue, version 2014.2.0.65). The superficial capillary plexus en-face image was automatically segmented with an inner boundary at 3 μm beneath the internal limiting membrane and an outer boundary set at 15 μm beneath the inner plexiform layer, while the deep capillary plexus en-face image had its inner boundary at 15 μm beneath the inner plexiform layer and an outer boundary at 70 μm beneath the inner plexiform layer. All scans were individually reviewed by one of the investigators (E.A.T.S.) for segmentation errors and repeated as necessary to allow sufficient image quality for post hoc analysis.

Table 2. Optical Coherence Tomography Angiography After Plaque Radiotherapy for Choroidal Melanoma (Tumor Features at Time of Plaque Radiotherapy)

Feature	Number (%), n = 65
Largest basal diameter, mean (median, range), mm	11 (11, 4–20)
Thickness, mean (median, range), mm	5 (5, 2–13)
Tumor quadrant	
Superior	12 (18)
Inferior	14 (22)
Nasal	15 (23)
Temporal	14 (22)
Macula	10 (15)
Distance to optic nerve, mean (median, range), mm	4 (3, 0–16)
Distance to foveola, mean (median, range), mm	4 (3, 0–18)
Pretreatment OCT features	
Macular subretinal fluid	23 (35)
Subfoveal fluid	21 (32)
Cystoid macular edema	6 (9)

Table 3. Optical Coherence Tomography Angiography After Plaque Radiotherapy for Choroidal Melanoma in 65 Consecutive Patients (Radiation Parameters)

Feature	Number (%), n = 65
Radiation dose, mean (median, range), cGy	
Tumor apex	7186 (7000, 6800–9316)
Tumor base	18,350 (15,090, 9063–47,890)
Foveola	5663 (3705, 547–28,140)
Optic disc	4036 (3346, 619–15,800)
Adjuvant treatment	
Transpupillary thermotherapy	28 (43)
Sector panretinal laser photocoagulation	41 (63)
Avastin prophylaxis q4 months	53 (82)
Follow-up after treatment, mean (median, range), months	46 (26, 4–216)
Ophthalmoscopically-evident radiation complications	
Radiation maculopathy present	40 (62)
Without OCT-evident macular edema	5 (8)
With OCT-evident macular edema	35 (54)
Radiation maculopathy absent	25 (38)
Without OCTA-evident macular ischemia	19 (29)
With OCTA-evident macular ischemia	6 (9)
Radiation papillopathy present	10 (15)
Radiation papillopathy absent	55 (85)

Foveal Avascular Zone Measurement

The size of the foveal avascular zone (FAZ) was assessed for all of the study images using Image J software (National Institutes of Health, Bethesda, MD). Before FAZ measurement, image scale was set using a known pixel distance of 304 pixels, 3-mm known anatomic distance, and a pixel aspect ratio of 1.0. This resulted in a scale of 101.33 pixels per mm. The largest and smallest FAZ diameters were subjectively identified, knowing that the normal FAZ was minimally irregular and variable in size.[15] The edge points were manually selected along the centerline of the vessels to determine FAZ area.

Capillary Density Measurement

The capillary density of the superficial and deep capillary plexus was measured separately in each 3 mm × 3 mm OCTA image obtained. Each scan was processed as a 304 × 304 pixel image, with a total pixel count of 92,416. The superficial capillary plexus and deep capillary plexus en-face images were both exported and opened in Adobe Photoshop CS3 (Adobe Systems Inc, San Jose, CA) for image processing without any postacquisition editing, and rendered using Adobe-RGB color scheme. Because flow above the decorrelation threshold will appear bright, vascular structures will appear bright whereas areas without perfusion will appear dark. Using the color range tool, the black (defined as RGB = 0-0-0) was selected and the inverse tool was used to select any color scheme other than black with a fuzziness factor greater than 140. Fuzziness factor was arbitrarily set to maintain consistency of measurement. The total number of pixels selected (×) was then expressed as a percentage (%) of the total number of pixels in the whole image (×/92,416 pixels per 3 × 3 mm scan).

Statistical Analysis

All recorded data and statistical analyses were performed using Microsoft Excel 2011. Fisher's exact test was used to compare categorical variables between

Table 4. Optical Coherence Tomography Angiography After Plaque Radiotherapy for Choroidal Melanoma (OCT and OCTA Features)

Feature	Irradiated Eyes, n = 65 (%)	Nonirradiated (Control) Eyes, n = 65 (%)	*P*
Visual acuity, mean (median, range),			
LogMAR	0.7 (0.5, 0–3)	0.1 (0, 0.1–0.3)	<0.001
Snellen equivalent	20/100 (20/60, 20/20-HM)	20/25 (20/20, 20/25–20/40)	
Central macular thickness on OCT, mean (median, range), μm	300 (277, 132–652)	271 (268, 235–339)	0.05
OCTA features			
Parafoveal nonperfusion, superficial plexus	19 (29)	0 (0)	<0.001
Parafoveal nonperfusion, deep plexus	20 (31)	0 (0)	<0.001
Telangiectasia, superficial plexus	3 (5)	1 (1)	0.62
Telangiectasia, deep plexus	2 (3)	1 (1)	1.00
Subretinal (type 2) neovascularization	0 (0)	0 (0)	1.00
Loss of choriocapillaries, within tumor margin	11 (17)	0 (0)	<0.001
Loss of choriocapillaries, outside tumor margin	0 (0)	0 (0)	1.00
Foveal avascular zone largest diameter, mean (median, range), mm			
Superficial plexus	1.185	0.669	<0.001
Deep plexus	1.455	0.894	<0.001
Foveal avascular zone smallest diameter, mean (median, range), mm			
Superficial plexus	0.647	0.517	0.01
Deep plexus	0.765	0.641	0.01
Foveal avascular zone area, mean (median, range), mm^2			
Superficial plexus	0.961	0.280	<0.001
Deep plexus	1.396	0.458	<0.001
Foveal capillary density, mean (median, range)			
Superficial plexus	21 (20, 2–42)	35 (37, 14–50)	<0.001
Deep plexus	17 (14, 2–53)	41 (42, 5–59)	<0.001

Reasons for inability to measure foveal avascular zone are severe macular ischemia >50% of parafovea (n = 5), severe cystoid macular edema causing artifactual enlargement of avascular zone (n = 5), macular elevation secondary to tumor with loss of OCT signal (n = 3). HM, hand motions.

irradiated eyes and the fellow nonirradiated (control) eyes, whereas paired Student's *t*-test was used to compare continuous data such as CMT, FAZ diameter, FAZ area, and capillary density. Parameters were also separately measured for the superficial and deep capillary plexus.

Results

There were 130 eyes in 65 patients in which 65 eyes had received plaque radiotherapy for choroidal melanoma and 65 eyes were contralateral, nonirradiated (control eyes), and included in the study. The patient demographics are listed in Table 1. The mean patient age at imaging was 55 years and most were female (39/65, 60%) and white (63/65, 98%). Medical history revealed diabetes mellitus in 4 patients (4/65, 4%) and hypertension in 25 (25/65, 38%), none with disease-related maculopathy.

The baseline tumor features are listed in Table 2. The mean largest tumor diameter was 11 mm, mean thickness was 5 mm, and most tumors were extramacular in location (55/65, 85%). The mean tumor distance to the optic nerve was 4 mm and mean distance to the foveola was 4 mm. There were 21 eyes (21/65, 32%) that had subfoveal fluid at baseline before plaque radiotherapy.

The radiation parameters are listed in Table 3. The mean tumor apex dose was 7,168 cGy and base dose was 18,350 cGy. The mean foveolar dose was 5,663 cGy and optic disc dose was 4,036 cGy. Adjuvant treatment for tumor consolidation included transpupillary thermotherapy (extramacular) in 28 eyes (28/65, 43%), and prevention of radiation retinopathy included sector panretinal photocoagulation in 41 eyes (41/65, 63%) and intravitreal bevacizumab prophylaxis in 53 eyes (53/65, 82%). At mean follow-up of 46 months post radiotherapy (median 26 months, range 4–216 months), clinically-evident radiation maculopathy by ophthalmoscopy was present in 40 eyes (62%) and absent in 25 (38%). Of these 25 eyes without radiation maculopathy, 6 demonstrated OCTA features of mild macular ischemia.

The OCT and OCTA features are listed in Table 4. A comparison (irradiated vs. nonirradiated [control] eye) revealed reduced mean logMAR visual acuity (0.7 vs. 0.1 [Snellen equivalent 20/100 vs. 20/25], $P < 0.001$), increased mean CMT (300 μm vs. 271 μm, $P = 0.05$), and significant findings of parafoveal nonperfusion in the superficial capillary plexus (29% vs. 0%, $P < 0.001$), parafoveal nonperfusion in the deep capillary plexus (31% vs. 0%, $P < 0.001$), and loss of choriocapillaris within the tumor margin (17% vs. 0%, $P < 0.001$). Subgroup analysis showed that

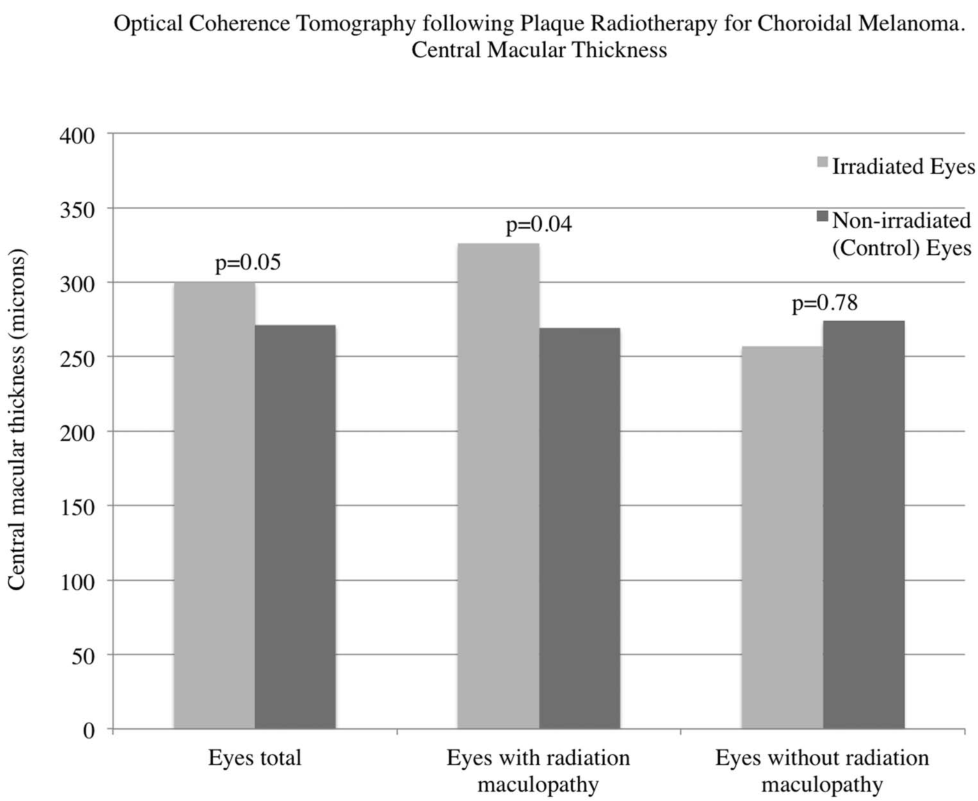

Fig. 1. Analysis of CMT in eyes treated with plaque radiotherapy for choroidal melanoma compared with the fellow (non-irradiated [control]) eye demonstrating significant difference of CMT in the entire group (eyes total) ($P = 0.05$) and in those with ophthalmoscopic evidence of radiation maculopathy ($P = 0.04$), but no difference in eyes without evidence of radiation maculopathy ($P = 0.78$).

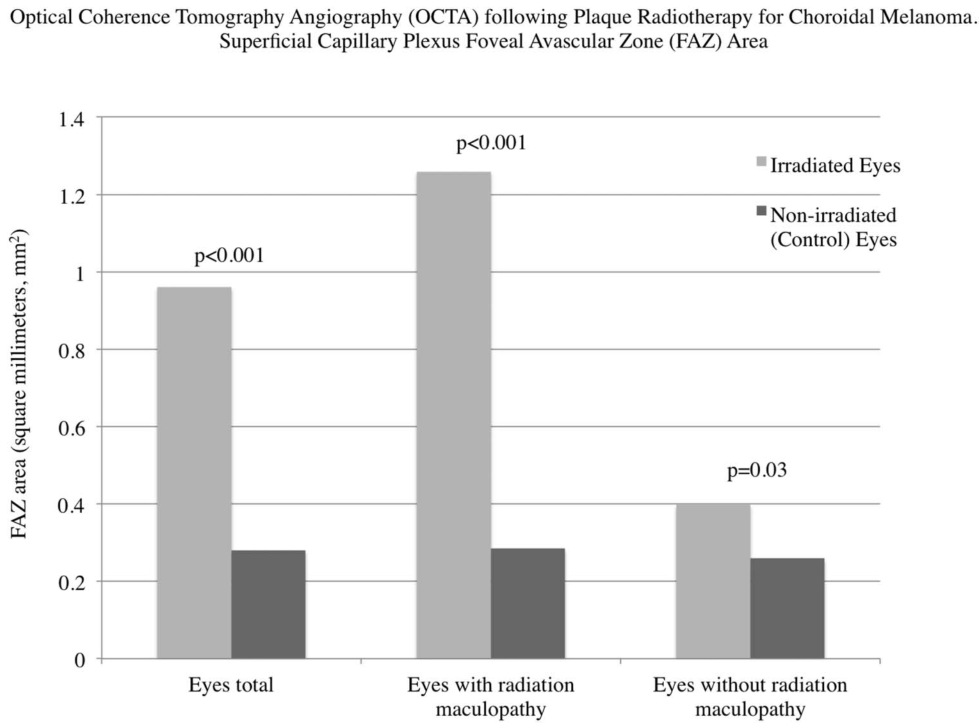

Fig. 2. Analysis of superficial capillary plexus FAZ area in eyes treated with plaque radiotherapy for choroidal melanoma compared with the fellow (nonirradiated [control]) eye demonstrating significant difference with enlargement of FAZ in the entire group (eyes total) ($P < 0.001$), in those with ophthalmoscopic evidence of radiation maculopathy ($P < 0.001$), and to a lesser extent those without evidence of radiation maculopathy ($P = 0.03$).

eyes with a history of clinically-evident radiation maculopathy had statistically significant increase in CMT ($P = 0.04$), whereas eyes with no clinically-evident radiation maculopathy had similar CMT as their fellow eye ($P = 0.78$) (Figure 1).

The FAZ was measured in diameter and area for both the superficial and deep capillary plexuses (Table 4). A comparison (irradiated vs. nonirradiated [control] eye) for the superficial capillary plexus of the mean largest FAZ diameter (1.185 vs. 0.669 mm,

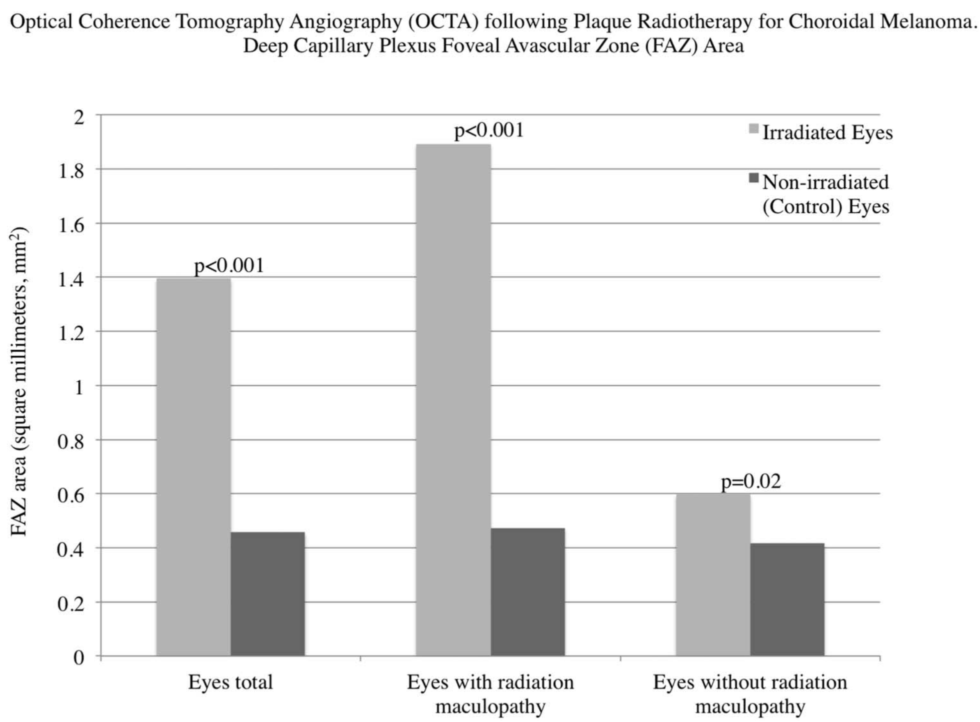

Fig. 3. Analysis of deep capillary plexus FAZ area in eyes treated with plaque radiotherapy for choroidal melanoma compared with the fellow (nonirradiated [control]) eye demonstrating significant difference with enlargement of FAZ in the entire group (eyes total) ($P < 0.001$), in those with ophthalmoscopic evidence of radiation maculopathy ($P < 0.001$), and to a lesser extent in those without evidence of radiation maculopathy ($P = 0.02$).

Table 5. Optical Coherence Tomography Angiography After Plaque Radiotherapy for Choroidal Melanoma in 65 Consecutive Patients

	Irradiated Eyes, n = 65 (%)					*P*	*P*
	Ophthalmoscopically-Evident Radiation Maculopathy, n = 40						
Feature	Active Radiation Maculopathy, n = 31	Resolved Radiation Maculopathy, n = 9*	Any Active or Resolved Radiation Maculopathy, n = 40	No Ophthalmoscopic Evidence of Radiation Maculopathy, n = 25	Nonirradiated (Control) Eyes, n = 65 (%)	Any Radiation Maculopathy (Irradiated Eyes vs. Control Eyes)	No Radiation Maculopathy (Irradiated Eyes vs. Control Eyes)
Ophthalmoscopic features of radiation maculopathy							
Retinal hemorrhage	15 (48)	0 (0)	15 (38)	0 (0)	0 (0)	<0.001	na
Retinal microaneurysm	6 (19)	0 (0)	6 (15)	0 (0)	0 (0)	0.002	na
Retinal exudation	9 (29)	0 (0)	9 (23)	0 (0)	0 (0)	<0.001	na
Retinal nerve fiber layer infarction	8 (25)	0 (0)	8 (20)	0 (0)	0 (0)	<0.001	na
Retinal cystoid edema	25 (81)	0 (0)	25 (63)	0 (0)	0 (0)	<0.001	na
Previous retinal cystoid edema	0 (0)	9 (100)	9 (23)	0 (0)	0 (0)	0.010	na
Central macular thickness, mean (median, range), μm	337 (301, 161–652)	286 (270, 202–387)	326 (301, 161–652)	257 (263, 132–427)	271 (268, 235–339)	0.01	0.16
Foveal avascular zone area, superficial plexus, mean (median, range), mm^2	1.383 (0.912, 0.355–3.892)	0.949 (0.521, 0.236–3.213)	1.259 (0.817, 0.236–3.892)	0.398 (0.335, 0.094–1.537)	0.278 (0.266, 0.068–0.739)	<0.001	0.03
Foveal avascular zone area, deep plexus, mean (median, range), mm^2	2.177 (1.963, 0.540–7.168)	1.184 (1.073, 0.230–3.487)	1.893 (1.259, 0.230–7.168)	0.597 (0.476, 0.099–1.665)	0.454 (0.434, 0.149–1.341)	<0.001	0.02
Foveal capillary density, superficial plexus, mean (median, range)	20 (19, 2–39)	19 (17, 9–41)	20 (18, 2–41)	24 (22, 14–43)	36 (37, 14–50)	<0.001	<0.001
Foveal capillary density, deep plexus, mean (median, range)	13 (11, 1–47)	15 (9, 1–37)	14 (10, 1–47)	22 (19, 7–53)	41 (42, 5–59)	<0.001	<0.001
Foveal nonperfusion, superficial plexus, mean (median, range)	12 (39)	3 (33)	15 (38)	4 (10)	0 (0)	<0.001	0.11

(*continued on next page*)

Table 5. (*Continued*)

	Irradiated Eyes, n = 65 (%)					*P*	*P*
	Ophthalmoscopically-Evident Radiation Maculopathy, n = 40						
Feature	Active Radiation Maculopathy, n = 31	Resolved Radiation Maculopathy, n = 9*	Any Active or Resolved Radiation Maculopathy, n = 40	No Ophthalmoscopic Evidence of Radiation Maculopathy, n = 25	Nonirradiated (Control) Eyes, n = 65 (%)	Any Radiation Maculopathy (Irradiated Eyes vs. Control Eyes)	No Radiation Maculopathy (Irradiated Eyes vs. Control Eyes)
Foveal nonperfusion, deep plexus, mean (median, range)	11 (35)	3 (33)	14 (35)	6 (15)	0 (0)	<0.001	0.02
Visual acuity, mean (median, range)						<0.001	<0.001
Logmar	1.2 (1, 0–3)	0.4 (0.4, 0.1–0.9)	1.0 (0.7, 0–3)	0.4 (0.3, 0–1.3)	0.1 (0, 0–0.3)		
Snellen equivalent	20/300 (20/200, 20/20-HM)	20/50 (20/50, 20/25–20/150)	20/200 (20/100, 20/20-HM)	20/50 (20/40, 20/20–20/400)	20/25 (20/20, 20/20–20/40)		
Follow-up after treatment, mean (median, range), months	58 (47, 7–216)	75 (76, 14–167)	62 (48, 7–216)	18 (17, 4–41)	45 (26, 3–216)	0.07	0.01

OCTA features based on presence or absence of clinically-evident radiation maculopathy.

*Resolved radiation maculopathy is defined as any prior clinical (hemorrhage, microaneurysms, exudates, nerve fiber layer infarcts) or OCT (cystoid macular edema) evident radiation maculopathy with complete absence of clinical and OCT evident signs at the time of OCTA analysis with or without treatment.

HM, hand motions; na, not applicable.

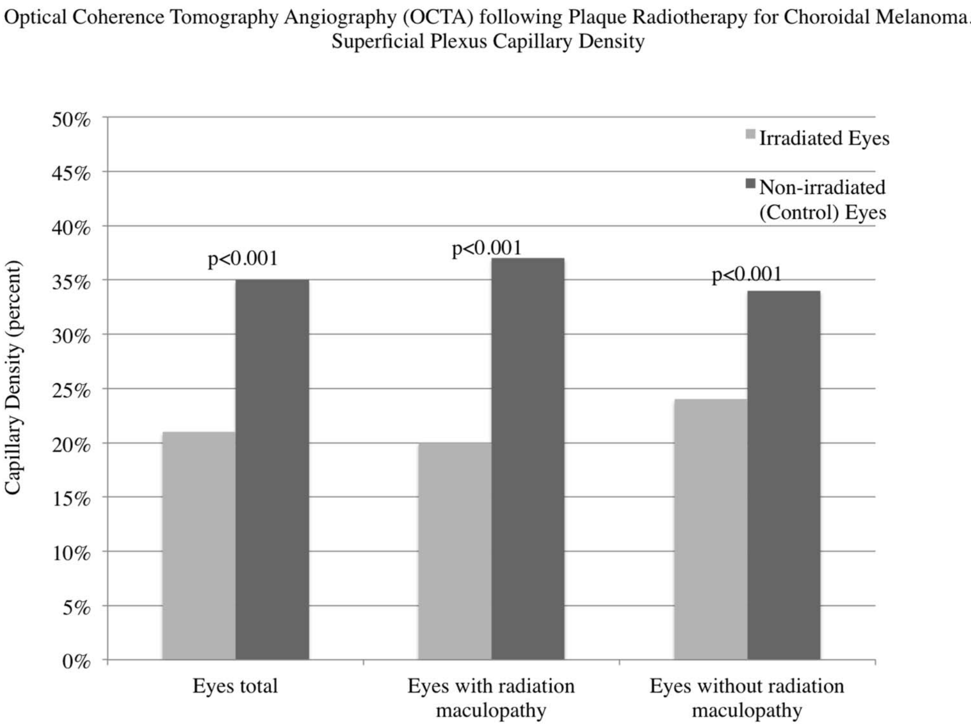

Fig. 4. Analysis of superficial plexus capillary density in eyes treated with plaque radiotherapy for choroidal melanoma compared with the fellow (nonirradiated [control]) eye demonstrating significant reduction in all eyes (eyes total) treated with plaque radiotherapy ($P < 0.001$) and even in eyes without any clinical evidence of radiation maculopathy ($P < 0.001$).

$P < 0.001$), mean smallest FAZ diameter (0.647 vs. 0.517 mm, $P = 0.01$), and mean FAZ area (0.961 vs. 0.280 mm^2, $P < 0.001$) were all significantly enlarged compared with their fellow nonirradiated (control) eyes (Figure 2). Evaluation of (irradiated vs. nonirradiated [control] eye) the deep capillary plexus of the mean largest FAZ diameter (1.455 vs. 0.894 mm, $P < 0.001$), mean smallest FAZ diameter (0.765 vs. 0.641 mm, $P = 0.01$), and mean FAZ area (1.396 vs. 0.458 mm^2, $P < 0.001$) all significantly enlarged

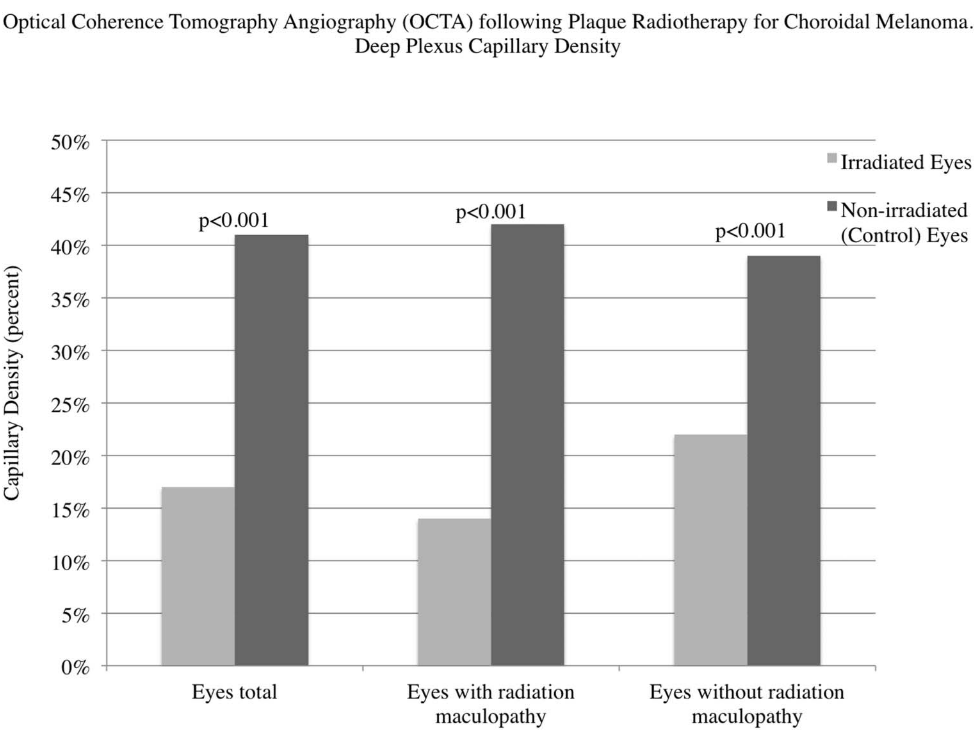

Fig. 5. Analysis of deep plexus capillary density in eyes treated with plaque radiotherapy for choroidal melanoma compared with the fellow (nonirradiated [control]) eye demonstrating significant reduction in all eyes (eyes total) treated with plaque radiotherapy ($P < 0.001$) and even in eyes without any clinical evidence of radiation maculopathy ($P < 0.001$).

compared with their fellow nonirradiated (control) eyes (Figure 3). Subgroup analysis showed that the enlargement of the FAZ area in the superficial (P = 0.03) and deep capillary plexus (P = 0.02) was maintained even in eyes without clinically-evident radiation maculopathy (Figures 2 and 3). There were 13 eyes excluded from FAZ measurements because of severe image distortion from advanced cystoid macular edema (5/65, 8%), extensive (>50%) macular nonperfusion (5/65, 8%), and macular elevation secondary to tumor (3/65, 5%).

The parafoveal capillary density (irradiated vs. nonirradiated [control] eye) in the superficial plexus (21% vs. 35%, $P < 0.001$) and deep plexus (17% vs. 41%, $P < 0.001$) was significantly decreased in irradiated eyes (Table 4). Subgroup analysis showed that the difference in parafoveal capillary density in the superficial (24% vs. 34%, $P < 0.001$) and deep (22% vs. 39%, $P < 0.001$) plexuses was preserved even in eyes without clinical maculopathy (Table 5, Figures 4 and 5).

A comparison of irradiated eyes with clinically-evident radiation maculopathy and those without radiation maculopathy versus control eyes is listed in Table 5. Findings revealed (clinically-evident radiation maculopathy vs. control eye) significantly increased mean CMT (326 vs. 271 μm, P = 0.01), increased FAZ in superficial (1.259 vs. 0.278 mm^2, $P <$ 0.001) and deep (1.893 vs. 0.454 mm^2, $P < 0.001$) plexuses, reduced capillary density in superficial (20% vs. 36%, $P < 0.001$) and deep (14% vs. 41%, $P <$ 0.001) plexuses, increased presence of foveal nonperfusion in superficial (38% vs. 0%, $P < 0.001$) and deep (35% vs. 0%, $P < 0.001$) plexuses, and reduced logMAR visual acuity (1.0 vs. 0.1 [Snellen equivalent 20/200 vs. 20/25], $P < 0.001$). Despite the lack of clinically-evident radiation maculopathy, findings revealed (clinically-absent radiation maculopathy vs. control eye) significantly increased FAZ in superficial (0.398 vs. 0.278 mm^2, P = 0.03) and deep (0.597 vs. 0.454 mm^2, P = 0.02) plexuses, reduced capillary density in superficial (24% vs. 36%, $P < 0.001$) and deep (22% vs. 41%, $P < 0.001$) plexuses, increased foveal nonperfusion in deep (15% vs. 0%, P = 0.02) plexus, and reduced logMAR visual acuity (0.4 vs. 0.1 [Snellen equivalent 20/50 vs 20/25], $P < 0.001$).

Discussion

Radiation maculopathy after plaque radiotherapy for choroidal melanoma can lead to profound vision loss and blindness.[9,18] In 2000, we studied visual results in 1,106 patients following plaque radiotherapy for choroidal melanoma and found reduction of vision by 5 Snellen lines or more in 33% at 5 years and 69% at 10 years, and with 5-year visual acuity of 20/200 or worse in 24% of eyes with small-size melanoma, 30% with

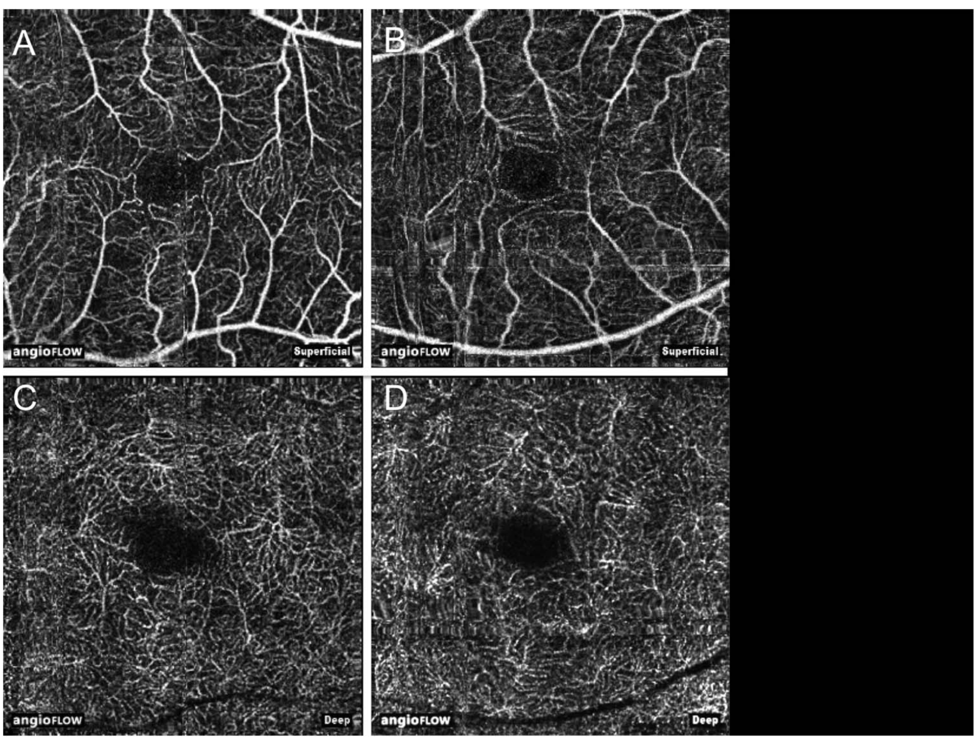

Fig. 6. Optical coherence tomography angiography of mild asymptomatic radiation maculopathy. After plaque radiotherapy of choroidal melanoma, there was no ophthalmoscopic evidence of radiation maculopathy. The superficial capillary plexus in the irradiated eye (**A**) showed slight enlargement of the FAZ, slight decrease in capillary density, particularly in the inferior hemisphere, and slight capillary telangiectasia compared with fellow, nonirradiated (control) eye (**B**). The mild enlargement of the FAZ area and loss of capillary density was also seen in deep capillary plexus (**C**) compared with the fellow, nonirradiated (control) eye (**D**).

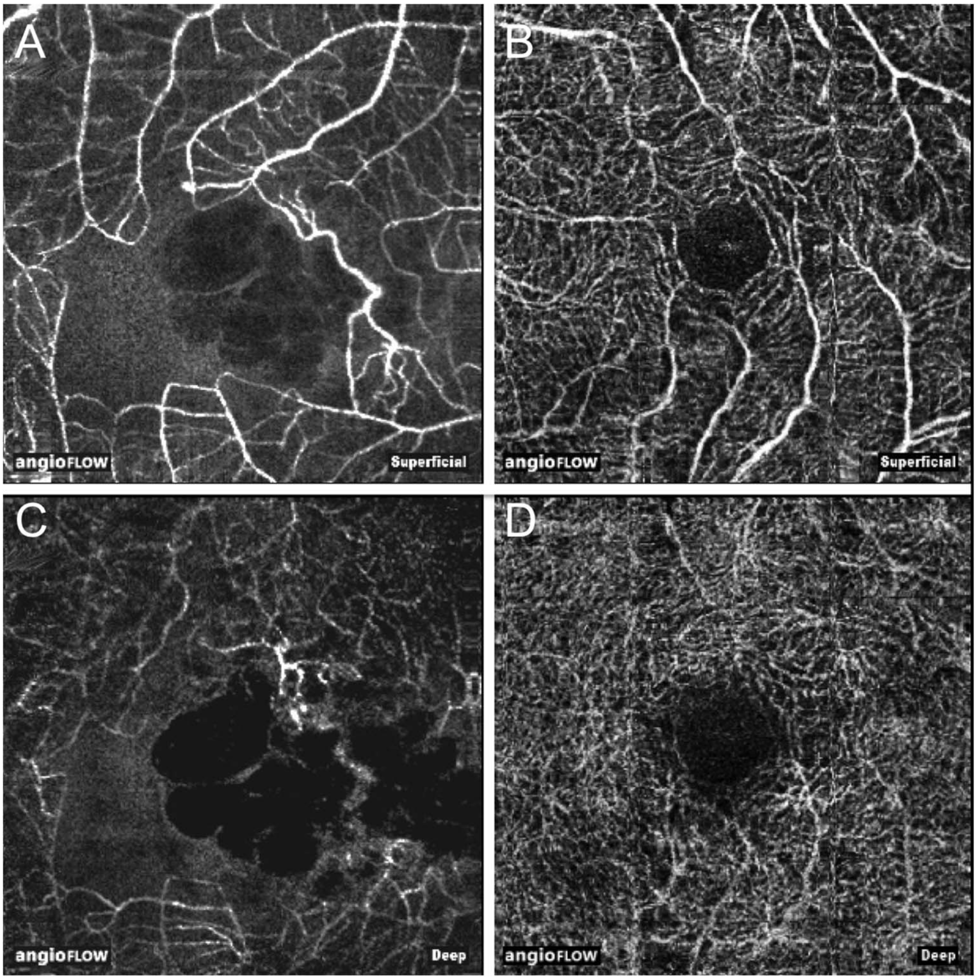

Fig. 7. Optical coherence tomography angiography of moderate symptomatic radiation maculopathy. After plaque radiotherapy of choroidal melanoma there was clinical evidence of radiation-related cystoid macular edema. The superficial capillary plexus in the irradiated eye (**A**) showed pronounced enlargement of the FAZ, vascular anastomosis in the nasal parafoveal region, microvascular dilation with telangiectasia, and decreased capillary density compared with the fellow, nonirradiated (control) eye (**B**). The deep capillary plexus (**C**) of the treated eye showed similar findings compared with the fellow, nonirradiated (control) eye (**D**).

medium-size melanoma, and 64% with large-size melanoma.[9] Similar results were recognized by the Collaborative Ocular Melanoma Study Group in which 6 lines of vision loss were detected in 49% at 3 years, and 43% of those with better than 20/200 at presentation eventually deteriorated to 20/200 or worse at 3 years.[18]

Attempts to treat radiation maculopathy have been made using focal laser photocoagulation of leaking macular vessels or nonperfused areas,[19] sector laser photocoagulation to site of radiation,[16,17] oral, subtenon's, or intravitreous corticosteroids,[20,21] and intravitreous injection of antivascular endothelial growth factor.[15,22–24] To some extent, favorable short-term results with mild visual recovery or preservation can be achieved with intravitreous medications, but durable effect has not been proven. Most protocols have depended on ophthalmoscopic evidence of radiation maculopathy or OCT evidence of macular edema. There are no therapeutic protocols based on OCTA-documented radiation maculopathy as this new technology has just recently become commercially available. In this analysis, we have found that OCTA is perhaps the most sensitive current method for detection of early radiation maculopathy, because parafoveal changes on OCTA were present in 62% of all eyes, even in eyes without clinically-evident maculopathy and those lacking OCT evidence of edema.

To ideally protect long-term visual acuity in eyes exposed to radiotherapy, there have been attempts at prevention of radiation complications.[7,15,23,24] In one study, we investigated prevention of radiation maculopathy using neoadjuvant intravitreal bevacizumab at the time of plaque removal and on a 4-month basis for 2 years.[15] We found moderately successful outcomes at 2 years (bevacizumab vs. control group) with reduction in clinically-evident radiation maculopathy (16% vs. 31%, $P = 0.001$), reduction in OCT-evident macular edema (26% vs. 40%, $P = 0.004$), reduction in moderate vision loss (33% vs. 57%, $P < 0.001$), and reduction in poor visual acuity (15% vs. 28%, $P = 0.004$).[15]

In this analysis, we demonstrated that early radiation maculopathy can be delineated on OCTA with quantitative and qualitative features. The quantitative features affected both the superficial and deep capillary plexuses and included irregular enlargement of the FAZ diameter and area, and reduction in measured parafoveal capillary density (Table 5). The qualitative features of early radiation maculopathy affected both the superficial and deep retinal capillary plexuses and included microvascular parafoveal nonperfusion and microvascular telangiectasia (Table 4, Figures 6 and 7). Optical coherence tomography angiography further demonstrated qualitative loss of choriocapillaris with loss of microvascularity within the tumor margin (Table 4). Even in eyes with no evidence of radiation maculopathy on ophthalmoscopy, there was significant reduction in parafoveal capillary density and enlargement of the FAZ in both superficial and deep capillary plexuses, leading to mild but significant reduction in visual acuity (Table 5, Figures 2–5). In contrast, eyes with no evidence of radiation maculopathy on ophthalmoscopy showed no difference in CMT on standard OCT analysis (Table 5, Figure 1). Based on these findings, we believe that OCTA is a remarkably sensitive technology for earliest detection of radiation maculopathy, often demonstrating subclinical microvascular ischemia. The findings on OCTA often precede the findings of edema on OCT and ophthalmoscopic findings of retinal telangiectasia, microaneurysm, exudation, hemorrhage, nerve fiber layer infarction, and retinal neovascularization.

Samara et al[14] studied OCTA in normal eyes and noted that there were variations in the FAZ area between superficial and deep plexuses and differences based on macular thickness. They observed that the FAZ was significantly larger in the deep plexus compared with the superficial plexus ($P < 0.0001$) and correlated inversely with CMT ($P < 0.0001$). There was no correlation with age or gender.[14] Similarly, in our series, we found that the deep capillary plexus showed larger mean FAZ diameter and area compared with superficial plexus in both irradiated and nonirradiated eyes (Table 4).

Limitations in this analysis include technical difficulty in obtaining reliable images from patients, distorted images in eyes with severe macular pathology, and image artifacts. In a few eyes, the macula was too severely edematous, nonperfused, or elevated from underlying tumor to allow reliable quantitative capillary density and FAZ measurements, despite our ability to provide a qualitative interpretation. Also, there is no current standard in measurement of capillary density, and previous quantification methods using histopathology, adaptive optics scanning laser ophthalmoscopy, and OCT have some variation in measured values, as well as representation.[25–32] However, assuming capillary density is statistically similar in both eyes of a single patient, we believe the decrease in capillary density is a true finding in eyes with radiation maculopathy. We believe our technique using a third party photo-editing software (Adobe Photoshop or ImageJ) can reliably quantify capillary density. The main benefit of using automated selection of vascularity is that it avoids human interpretation bias; however, this technique is limited by the fact that image artifacts (blink lines, out of focus, or segmentation errors) can appear to reduce capillary density. Considering the speed of current OCTA machines, these artifacts are commonly present and it is nearly impossible to avoid artifacts with current technique. Improvements in technology and image capturing techniques can reduce this limitation, and further studies could verify our findings.

Future studies on this topic should focus on the precise timing and evolution of OCTA microvascular changes based on serial monitoring of eyes before and after radiotherapy. Correlation of microvascular foveal damage with subretinal fluid, tumor location and size, and radiation parameters could be valuable. Understanding the initial microvascular process of radiation maculopathy could lead to modification of treatments for visual preservation.

Key words: eye, choroid, tumor, melanoma, plaque radiotherapy, radiation maculopathy, optical coherence tomography angiography, OCTA, macula, fovea, foveal avascular zone.

References

1. Shields JA, Shields CL. Management of posterior uveal melanoma. Past, present and future. The 2014 Charles L. Schepens Lecture. Ophthalmology 2015;122:414–428.
2. Shields CL, Shields JA. Recent developments in the management of choroidal melanoma. Curr Opin Ophthalmol 2004;15: 244–251.
3. Shields JA, Shields CL. Intraocular Tumors. An Atlas and Textbook. 3rd ed. Philadelphia, PA: Lippincott Wolters Kluwers; 2016:155–191.
4. Gunduz K, Shields CL, Shields JA, et al. Radiation retinopathy following plaque radiotherapy of posterior uveal melanoma. Arch Ophthalmol 1999;117:609–614.
5. Bianciotto C, Pirondini C, Shields CL, et al. Proliferative radiation retinopathy following plaque radiotherapy for uveal melanoma. Ophthalmology 2010;117:1005–1012.
6. Horgan N, Shields CL, Mashayekhi A, et al. Early macular morphological changes following plaque radiotherapy for uveal melanoma. Retina 2008;28:263–273.
7. Horgan N, Shields CL, Mashayekhi A, Shields JA. Classification and treatment of radiation maculopathy. Curr Opin Ophthalmol 2010;21:233–238.

8. Gunduz K, Shields CL. Radiation retinopathy and papillopathy. In: Yanoff M, Duker J, eds, Ophthalmology. 4th ed. Philadelphia, PA: Elsevier Saunders Co; 2014:565–569.
9. Shields CL, Shields JA, Cater J, et al. Plaque radiotherapy for uveal melanoma. Long-term visual outcome in 1106 patients. Arch Ophthalmol 2000;118:1219–1228.
10. Mashayekhi A, Schonbach E, Shields JA, Shields CL. Early subclinical macular edema in eyes with uveal melanoma. Association with future cystoid macular edema following plaque radiotherapy. Ophthalmology 2015;122:1023–1029.
11. Spaide RF, Klancnik JM, Cooney MJ. Retinal vascular layers imaged by fluorescein angiography and optical coherence tomography angiography. JAMA Ophthalmol 2015;133:45–50.
12. De Carlo TE, Roman A, Waheed NK, Duker JS. A review of optical coherence tomography angiography (OCTA). Int J Retina Vitreous 2015;1:e1–e15.
13. Nagiel A, Sadda SVR, Sarraf D. A promising future for optical coherence tomography angiography. JAMA Ophthalmol 2015; 133:629–630.
14. Samara W, Say EAT, Khoo CTL, et al. Correlation of foveal avascular zone size with foveal morphology in normal eyes using optical coherence tomography angiography. Retina 2015;35:2188–2195.
15. Shah SU, Shields CL, Bianciotto CG, et al. Intravitreal bevacizumab injection at 4-month intervals for prevention of macular edema following plaque radiotherapy of uveal melanoma. Ophthalmology 2014;121:269–275.
16. Finger PT, Kurli M. Laser photocoagulation for radiation retinopathy after ophthalmic plaque radiation therapy. Br J Ophthalmol 2005;89:730–738.
17. Materin MA, Bianciotto C, Wu C, Shields CL. Sector laser photocoagulation for the prevention of macular edema following plaque radiotherapy for uveal melanoma. Retina 2012;32:1601–1607.
18. Melia BM, Abramson DH, Albert DM, et al; Collaborative Ocular Melanoma Study. Collaborative Ocular Melanoma Study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma. I. Visual acuity after 3 years. COMS report no. 16. Ophthalmology 2001;108:348–366.
19. Hykin PG, Shields CL, Shields JA, Arevalo JF. The efficacy of focal laser therapy in the management of radiation induced macular edema. Ophthalmology 1998;105:1425–1429.
20. Horgan N, Shields CL, Mashayekhi A, et al. Periocular triamcinolone for prevention of macular edema following plaque radiotherapy of uveal melanoma. Retina 2008;28: 987–995.
21. Shields CL, Demirci H, Dai V, et al. Intravitreal triamcinolone acetonide for radiation maculopathy following plaque radiotherapy for choroidal melanoma. Retina 2005;25:868–874.
22. Mashayekhi A, Rojanaporn D, Al-Dahmash S, et al. Monthly intravitreal bevacizumab for macular edema after iodine-125 plaque radiotherapy of uveal melanoma. Eur J Ophthalmol 2014;24:228–234.
23. Wen JC, McCannel TA. Treatment of radiation retinopathy follow plaque brachytherapy for choroidal melanoma. Curr Opin Ophthalmol 2009;20:200–204.
24. Reichstein D. Current treatments and preventive strategies for radiation retinopathy. Curr Opin Ophthalmol 2015;26: 157–166.
25. Tan PE, Yu PK, Balaratnasingam C, et al. Quantitative confocal imaging of the retinal microvasculature in the human retina. Invest Ophthalmol Vis Sci 2012;53:5728–5736.
26. Mendis KR, Balaratnasingam C, Yu P, et al. Correlation of histologic and clinical images to determine the diagnostic value of fluorescein angiography for studying retinal capillary detail. Invest Ophthalmol Vis Sci 2010;51:5864–5869.
27. Tam J, Martin JA, Roorda A. Noninvasive visualization of parafoveal capillaries in humans. Invest Ophthalmol Vis Sci 2010;51:1691–1698.
28. Popovic Z, Knutsson P, Thaung J, et al. Noninvasive imaging of human foveal capillary network using dual-conjugate adaptive optics. Invest Ophthalmol Vis Sci 2011;52:2649–2655.
29. Chan G, Balaratnasingam C, Xu J, et al. In vivo optical imaging of human retinal capillary networks using speckle variance optical coherence tomography with quantitative clinic-histological correlation. Microvasc Res 2015;100:32–39.
30. Pinhas A, Razeen M, Dubow M, et al. Assessment of perfused foveal microvascular density and identification of nonperfused capillaries in healthy and vasculopathic eyes. Invest Ophthalmol Vis Sci 2014;55:8056–8066.
31. Jia Y, Bailey S, Hwang TS, et al. Quantitative optical coherence tomography angiography of vascular abnormalities in the living human eye. Proc Natl Acad Sci U S A 2015;112:E2395–E2402.
32. Kuehlewein L, Tepelus TC, An L, et al. Noninvasive visualization and analysis of the human parafoveal capillary network using swept source OCT optical microangiography. Invest Ophthalmol Vis Sci 2015;56:3984–3988.